

Climate Change: Causes, Impacts, and Solutions

Introduction

Climate change refers to long-term shifts in global temperatures and weather patterns. While climate has always varied naturally, since the mid-20th century human activities have been the primary driver of accelerating change, particularly through the burning of fossil fuels.

The scientific consensus on human-caused climate change is overwhelming. Thousands of peer-reviewed studies and reports from major scientific institutions including NASA, NOAA, and the IPCC confirm that Earth is warming at an unprecedented rate driven by human greenhouse gas emissions.

Main Causes

Burning fossil fuels — coal, oil, and natural gas — for energy and transportation releases carbon dioxide (CO₂) and other greenhouse gases into the atmosphere. This is the single largest contributor to human-caused climate change, accounting for roughly 75% of global greenhouse gas emissions.

Deforestation contributes to climate change in two ways: trees that are cut or burned release the carbon they have stored, and the loss of forests reduces Earth's capacity to absorb CO₂ from the atmosphere. Tropical deforestation accounts for approximately 10% of global emissions.

Agriculture contributes through methane emissions from livestock digestion and rice paddies, nitrous oxide from fertilizer use, and deforestation for farmland. The food system as a whole accounts for roughly one-quarter of global greenhouse gas emissions.

Key Impacts

Rising temperatures are the most direct impact. Global average surface temperature has already increased by approximately 1.1 degrees Celsius above pre-industrial levels. This warming is not uniform — the Arctic is warming roughly four times faster than the global average.

Sea level rise threatens coastal communities worldwide. Oceans have risen by about 20 cm since 1900 and are rising at an accelerating rate due to thermal expansion of warmer water and melting of glaciers and ice sheets. Low-lying island nations and coastal megacities face existential risks.

Extreme weather events are becoming more frequent and intense. Climate change increases the intensity of hurricanes, the frequency and severity of droughts, the extent of wildfires, and the magnitude of flooding events. Attribution science can now quantify climate change's contribution to specific extreme events.

The Paris Agreement

The Paris Agreement, adopted in December 2015 by 196 parties, is a legally binding international treaty on climate change. Its central aim is to limit global warming to well below 2 degrees Celsius above pre-industrial levels, with efforts to limit warming to 1.5 degrees Celsius.

Each signatory country submits Nationally Determined Contributions (NDCs) — commitments to reduce emissions and adapt to climate impacts. These NDCs are reviewed and updated every five years, with the expectation that ambition will increase over time.

Current NDCs, if fully implemented, would still lead to approximately 2.5-3 degrees Celsius of warming by 2100. Closing the gap between current commitments and the 1.5-degree target requires rapid and far-reaching transformations across energy, land use, transport, and industry.

Solutions

Renewable energy transition is the cornerstone of climate mitigation. Solar and wind power costs have fallen by 90% and 70% respectively over the past decade, making clean electricity the cheapest source of new power generation in most of the world.

Energy efficiency improvements in buildings, industry, and transportation can reduce emissions while often saving money. Building retrofits, efficient appliances, industrial process improvements, and vehicle electrification all contribute significantly to emissions reductions.

Carbon dioxide removal (CDR) technologies aim to actively pull CO₂ from the atmosphere. Options range from nature-based solutions like reforestation and soil carbon sequestration to engineered approaches like direct air capture. Most climate scenarios require some level of CDR alongside steep emissions reductions.